SUPPLY CHAIN INTELLIGENCE

Titanium Models A and B

End-to-end data, training and performance report

A complete record of the data lineage, preprocessing, engineered features, training design, model selection, validation, disruption testing, post-disruption monitoring, results and remaining limitations for the titanium import-price and supplier-shift models.

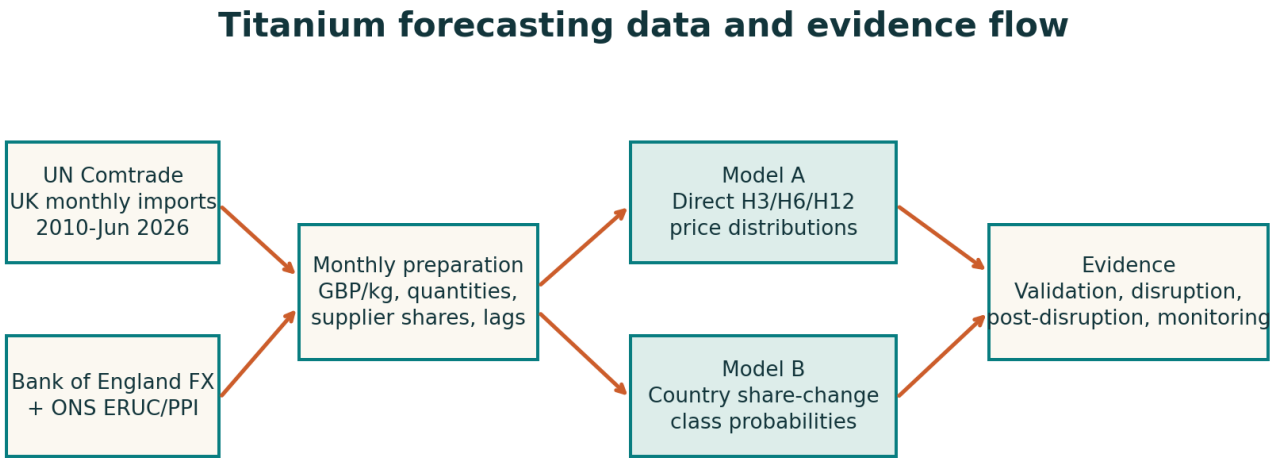
MODEL A	MODEL B
Import-price forecast	Country supply-shift classification
Direct H3, H6 and H12 P05/P50/P95	Collapse, fall, stable and rise probabilities
6/6 validation configurations pass	Beats baselines; calibration remains limited

Prepared 6 October 2026

Evidence current through June 2026

Executive summary

The two models share a data foundation but solve different prediction tasks.



Model A forecasts titanium import unit value in GBP/kg for HS 810820 and HS 810890. It produces direct P05, P50 and P95 forecasts at 3, 6 and 12 months. All six selected HS/horizon configurations passed blocked validation. All four primary H3/H6 combinations also passed the 2022-2023 disruption test. Post-disruption evidence is mixed: three of four primary combinations pass, while HS 810890 H3 narrowly loses to the no-change baseline.

Model B predicts whether each supplier's UK import-weight share will collapse, fall, remain stable or rise. H3 Random Forest and H6 Extra Trees beat both probability baselines and pass calibration during the disruption test. Validation and post-disruption probabilities still fail the strict maximum calibration-gap rule. H12 is reporting-only.

Question	Model A	Model B
Prediction	Future GBP/kg price distribution	Future supplier-share movement class
Unit	GBP per kg	Probability for four classes
Horizon	Direct H3, H6, H12	Direct H3, H6, H12
Primary demo status	Suitable research demonstration	Directional research demonstration
Production status	Not approved	Not approved

1. Data collection and provenance

The implementation uses three permitted source families and explicitly excludes unrelated or incomplete sources.

Source	Local input	Coverage	Use
UN Comtrade	17 annual TradeData CSV exports	Jan 2010-Jun 2026	UK monthly imports, USD value, net weight, partner rows and supplier shares
Bank of England	Bank of England Database.csv	Jan 2010-Jun 2026	Monthly USD per GBP; converts USD trade value and unit value into GBP
ONS ERUC/PPI	Inflation Data/ppi.csv; ERUC series	Jan 2010-Jun 2026	Import-price inflation proxy for other non-ferrous metals, 2015=100

UN Comtrade extraction: 28 September 2026. Reporter is the United Kingdom, trade flow is imports, frequency is monthly, and the retained as-reported HS codes are 810820 and 810890.

HS 810820: titanium unwrought and powders. **HS 810890:** titanium bars, sheet, forgings and other articles. HS 810830 waste and scrap is excluded.

Excluded sources: Metals-API titanium benchmark data, USGS titanium sponge data, and World Bank aluminium/nickel prices are not model inputs. Their shorter, stale or cross-metal histories did not meet the final source policy.

The ONS ERUC series is titled 'PPI INDEX IMPORT - C2445 Other non-ferrous metal 2015=100'. It is a broad non-ferrous inflation proxy, not a titanium-only market benchmark.

Availability assumptions prevent look-ahead: UN Comtrade month-end plus 42 days, Bank of England month-end plus 7 days, and ERUC month-end plus 31 days.

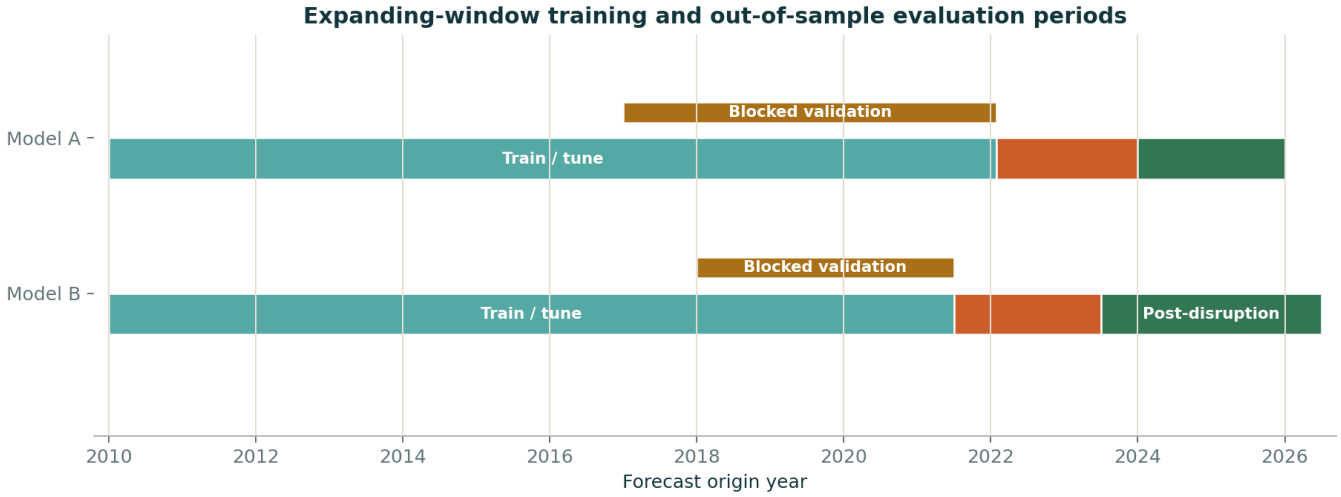


Figure 1. Expanding-window model development and evaluation periods. Validation overlaps the earlier model-selection history but uses only prior data within each blocked fold.

2. Data preparation and quality controls

The pipeline preserves source anomalies as flags and prevents future labels from entering historical training windows.

Step	Operation	Control
1	Load all annual Comtrade CSVs with Latin-1 encoding	Retain UK monthly imports only
2	Keep HS 810820 and 810890; preserve World and partner rows	One World row per HS/month
3	Use World primaryValue and positive World net weight	Reconcile World value to partner sum
4	Reconstruct missing/zero World weights from partner-country weights	37 rows flagged; values are not silently replaced
5	Compute USD/kg, join monthly FX, then compute GBP/kg	GBP value = USD value / USD-per-GBP
6	Join ERUC and create 1/3/6/12-month changes	Availability-date checks before use
7	Create price, quantity, volatility, supplier and seasonal features	All lags use current or earlier months
8	Create direct future H3/H6/H12 labels	Target availability date controls training eligibility
9	Apply data-quality and target-reliability weights	Problematic targets are downweighted, not hidden

The monthly GBP unit value is calculated as **trade value in GBP / net weight in kg**. It is an import unit value, not a quoted exchange price. Mix changes across grades, forms and suppliers can therefore move the value even when the underlying commodity market is stable.

Profile	Model A matrix	Model B panel
Shape	396 rows x 134 columns	1,188 rows x 75 columns
Grain	One month per HS code	One month per supplier group
Entities	2 HS products	6 supplier groups
Time	Jan 2010-Jun 2026	Jan 2010-Jun 2026
Feature-complete	372/396	1116/1188
Training-eligible H3/H6/H12	354 / 348 / 338	1098 / 1080 / 1044

Missing cells are expected at the edges of the series. The first 12 months lack some lagged features, while the final 3, 6 or 12 months lack future target labels. These rows are excluded horizon-by-horizon rather than imputed as outcomes. FX and ERUC joined successfully for every HS/month row.

Quality condition	Count	Treatment
Reconstructed World weight	37 HS-month rows	Partner weights summed; explicit flag retained
UN-estimated World weight	91 HS-month rows	Flag retained as model-quality information
Robust unit-value outlier	6 HS-month rows	Preserved and excluded/weighted where required
Partner anomaly	13 HS-month rows	Review flag retained
Target reliability	310 reliable; 79 caution; 7 review	Quality-weighted fitting

3. Feature engineering

Both models use only information assumed available at the simulated forecast date.

Feature family	Model A price forecast	Model B supply-shift forecast
Price history	GBP price changes 1/2/3/6/12m; 6/12m volatility	Combined titanium unit-value changes 1/3/6/12m
Import volume	Quantity changes 1/3/12m; 6/12m volatility; log weight; low-volume ratio/flag	Total import changes 1/3/6/12m; country log quantity and changes
Supplier structure	Six monthly supplier shares; HHI; top share; 3m concentration changes	Country share and changes 1/3/6/12m; HHI; top share; country identity
External context	FX changes/volatility; ERUC changes/volatility	FX changes and ERUC changes 1/3/6/12m
Calendar	Month sine/cosine; January 2021 EU collection-change flag	Month sine/cosine; January 2021 EU collection-change flag
Quality	Price shock, low volume and UN-estimated weight flags	Reconstructed/estimated weight flags and partner count

Log changes express proportional movement: log(value[t] / value[t-k]). Volatility is the rolling standard deviation of historical changes. Month sine and cosine encode the annual cycle without treating December and January as far apart.

Future target columns, target months, availability dates, identifiers and audit flags are retained for traceability but are never passed to the estimator as predictive inputs.

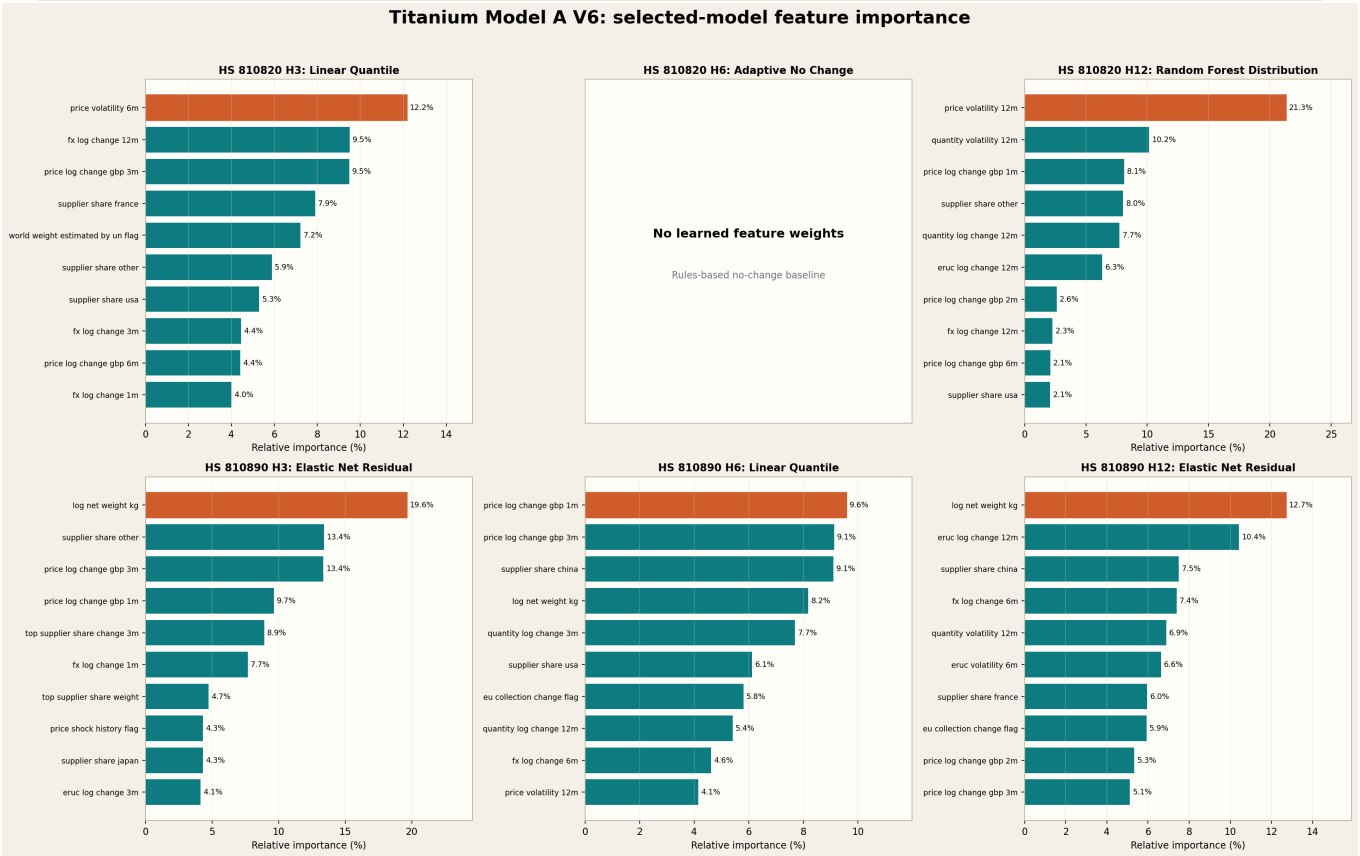


Figure 2. Model A selected-model feature influence. Importance describes model reliance, not causation.

4. Model A: titanium import-price forecast

Model A predicts a full future price distribution separately for each HS product and horizon.

Target: future logarithmic change in monthly import unit value in GBP/kg. For horizon h: log(price[t+h]) - log(price[t]). The predicted future price is price[t] multiplied by exp(predicted log change).

Outputs: P05, P10, P25, P50, P75, P90 and P95 price quantiles. The dashboard focuses on P05, P50 and P95. H3, H6 and H12 are direct targets; they are not recursive H1 forecasts.

Separate models: HS 810820 and HS 810890 are not averaged because their unit values and market composition differ materially.

Candidate	Role
Linear quantile	Fits conditional price-change quantiles directly
Gradient boosting quantile	Nonlinear tree boosting for each quantile
Random forest distribution	Uses the distribution of tree predictions
Elastic Net residual	Regularized linear correction around a base forecast
Huber residual	Robust linear residual model, less sensitive to outliers
Adaptive no-change	Safe history-based forecast with adaptive interval
No-change baseline	Reference model: future price equals current price

Candidate selection uses blocked validation ending January 2022. A candidate is eligible when calibrated P05-P95 coverage is at least 90% and Mean Interval Score is lower than the independently calibrated no-change baseline. The minimum eligible MIS wins; raw pinball loss is only a tie-breaker.

HS	H	Selected model	Train rows	Coverage	MIS ratio
810820	H3	linear_quantile	181	93.3%	0.970
810820	H6	adaptive_no_change	178	93.3%	0.978
810820	H12	random_forest_distribution	172	93.3%	0.774
810890	H3	elastic_net_residual	173	93.3%	0.869
810890	H6	linear_quantile	170	93.3%	0.796
810890	H12	elastic_net_residual	166	93.3%	0.800

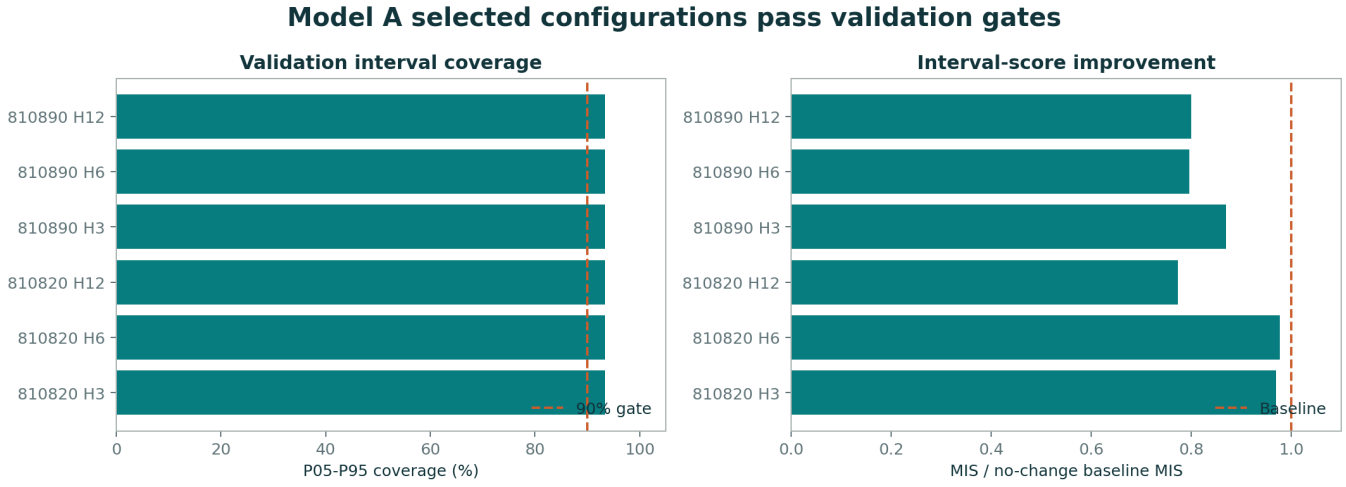


Figure 3. Every selected Model A configuration clears both validation interval gates.

5. Model A evaluation results

Point accuracy and interval quality are evaluated together; interval gates drive acceptance.

Metric	Meaning
MAE / RMSE	Average and squared-error-sensitive P50 price error in GBP/kg
sMAPE	Scale-normalized percentage difference between actual and P50
Pinball loss	Quantile forecast error; lower is better
90% coverage	Share of actual outcomes inside P05-P95; gate is at least 90%
MIS	Interval width plus a severe penalty for misses; lower is better
MIS ratio	Selected-model MIS divided by no-change baseline MIS; pass is below 1

Validation: forecast origins January 2017-January 2022

HS	H	Model	MAE	sMAPE	Coverage	MIS ratio	Result
810820	H3	linear_quantile	GBP 1.08/kg	13.7%	93.3%	0.970	Pass
810820	H6	adaptive_no_change	GBP 1.20/kg	14.8%	93.3%	0.978	Pass
810820	H12	random_forest_distribution	GBP 1.39/kg	16.3%	93.3%	0.774	Pass
810890	H3	elastic_net_residual	GBP 9.36/kg	23.8%	93.3%	0.869	Pass
810890	H6	linear_quantile	GBP 11.17/kg	28.0%	93.3%	0.796	Pass
810890	H12	elastic_net_residual	GBP 14.07/kg	32.4%	93.3%	0.800	Pass

HS 810820 validation sMAPE ranges from 13.7% to 16.3%. HS 810890 is more difficult, with 23.8% to 32.4% sMAPE and much larger GBP/kg errors because that product series is higher-priced and more volatile.

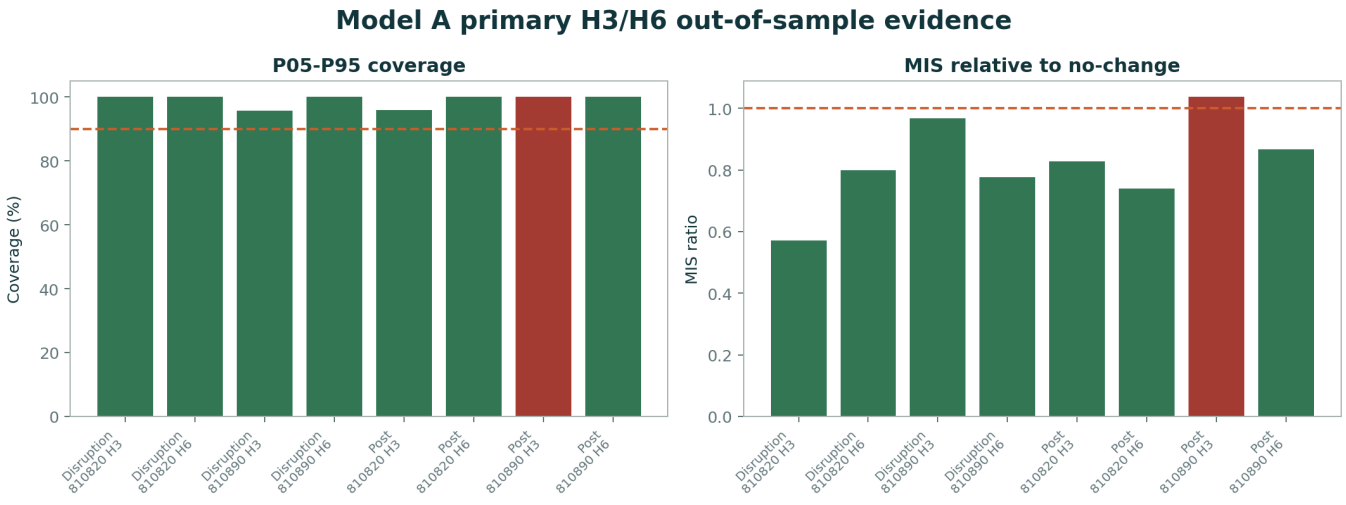


Figure 4. Primary H3/H6 disruption and post-disruption interval tests.

Period	HS	H	MAE	sMAPE	Coverage	MIS ratio	Result
Disruption	810820	H3	GBP 0.88/kg	8.4%	100.0%	0.572	Pass
Disruption	810820	H6	GBP 1.40/kg	13.2%	100.0%	0.799	Pass
Disruption	810890	H3	GBP 7.60/kg	16.9%	95.7%	0.967	Pass
Disruption	810890	H6	GBP 8.24/kg	18.0%	100.0%	0.777	Pass

Period	HS	H	MAE	sMAPE	Coverage	MIS ratio	Result
Post	810820	H3	GBP 1.33/kg	10.6%	95.8%	0.828	Pass
Post	810820	H6	GBP 0.91/kg	7.9%	100.0%	0.740	Pass
Post	810890	H3	GBP 4.12/kg	9.4%	100.0%	1.037	Review
Post	810890	H6	GBP 6.50/kg	14.4%	100.0%	0.867	Pass

The disruption test covers February 2022-December 2023 and all four primary H3/H6 combinations pass. The post-disruption evaluation covers January 2024-December 2025: both HS 810820 horizons and HS 810890 H6 pass; HS 810890 H3 has 100% coverage but an MIS ratio of 1.037, so the no-change baseline is slightly sharper.

H12 is reported but not counted in the four primary combinations. Post-disruption H12 has only 18 of 24 outcomes because the source ends in June 2026. Its 75% observed coverage and infinite MIS are therefore incomplete evidence, not a valid acceptance result.

6. Model B: titanium supply shift by country

Model B predicts how each supplier's share of UK titanium import weight will change.

The six supplier groups are USA, Japan, France, Russian Federation, China and Other countries. Monthly country weight is reconstructed from each HS row's supplier share multiplied by net weight, then aggregated across HS 810820 and 810890.

For horizon h, the numeric target is share[t+h] - share[t]. Fixed class boundaries are derived only from pooled H3 changes ending June 2021.

Class	Future share change
Collapse	Below -14.4608 percentage points
Fall	-14.4608 to below -2.2945 percentage points
Stable	-2.2945 through +2.0887 percentage points
Rise	Above +2.0887 percentage points

The specification originally requested class-boundary history from 2005. Available monthly data begins in 2010, so the frozen thresholds use January 2010-June 2021. This is a documented specification deviation.

Horizon	Selected model	Parameters	Temperature	Final train rows	Use
H3	random_forest	n_estimators=140, max_depth=8, min_samples_leaf=4	1.15	1098	Primary
H6	extra_trees	n_estimators=160, max_depth=10, min_samples_leaf=3	0.75	1080	Primary
H12	random_forest	n_estimators=140, max_depth=8, min_samples_leaf=4	1.35	1044	Reporting only

Candidate families are multinomial logistic regression, Random Forest and Extra Trees. Numeric values receive median imputation. Logistic regression also receives standardization. Country identity is one-hot encoded. Probability temperature is selected on blocked validation to reduce Brier score.

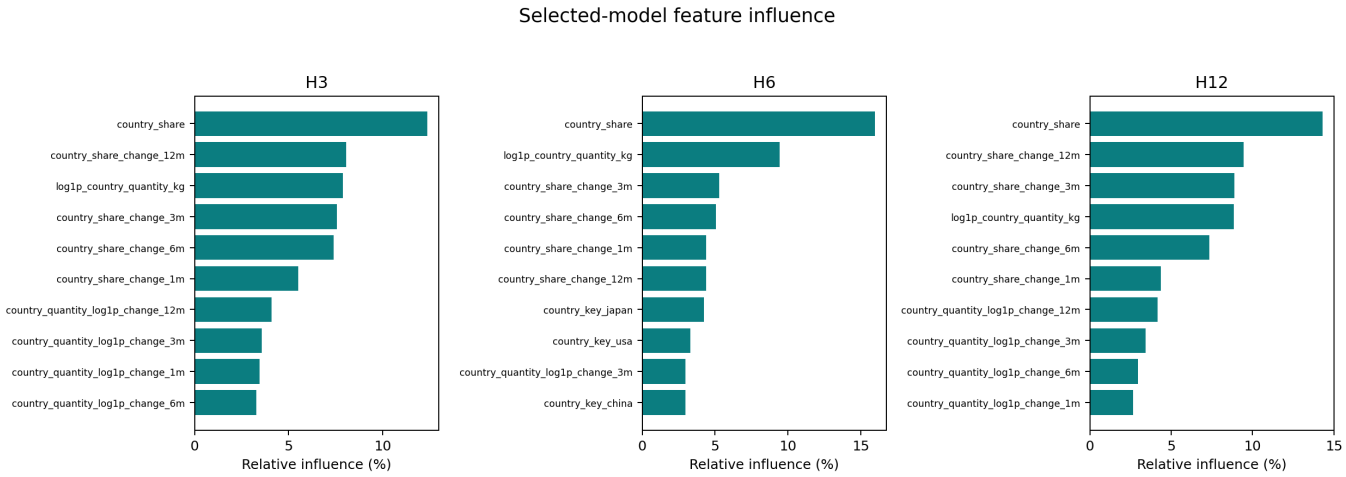


Figure 5. Model B feature importance. Current country share and its recent changes dominate all horizons.

7. Model B evaluation results

Probability accuracy beats simple baselines, but reliability varies by period and horizon.

Metric	Meaning
Accuracy	Share of rows whose most probable class equals the actual class
Balanced accuracy	Average class recall; reduces dominance by common classes
Macro F1	Equal-weight average classification F1 across four classes
Brier score	Squared error of the four predicted probabilities; lower is better
Log loss	Penalty for assigning low probability to the true class; lower is better
ECE	Weighted average difference between confidence and observed accuracy
Maximum calibration gap	Worst populated reliability-bin gap; required at or below 10%

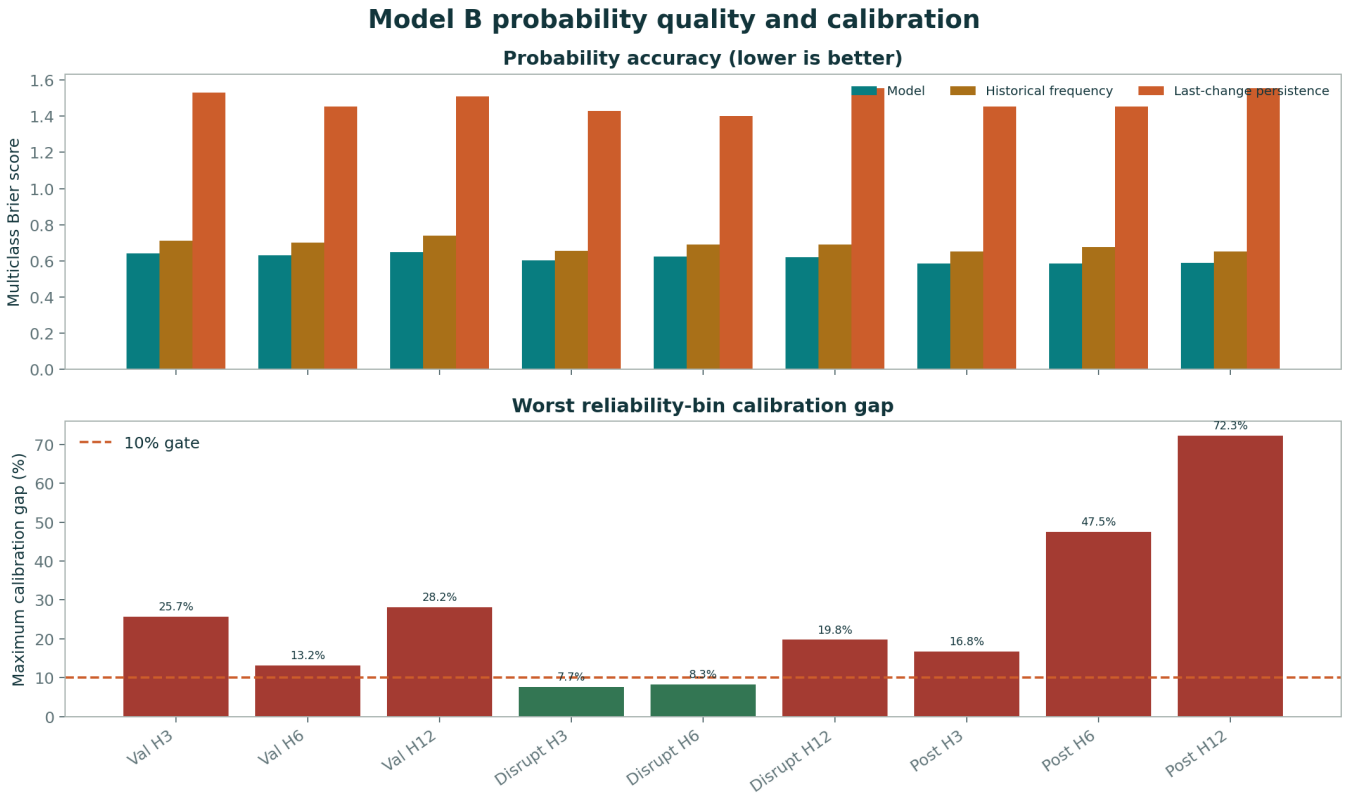


Figure 6. Model B beats both Brier baselines in every shown split, but only disruption H3/H6 meet the strict maximum-gap calibration rule.

Period	H	Model	Accuracy	Bal. acc.	Macro F1	Brier	Max gap	Result
Disruption	H3	random_forest	50.0%	52.4%	51.2%	0.603	7.7%	Pass
Disruption	H6	extra_trees	47.2%	53.3%	47.9%	0.624	8.3%	Pass
Disruption	H12	random_forest	45.1%	47.0%	45.0%	0.621	19.8%	Report
Post	H3	random_forest	56.1%	54.5%	52.5%	0.583	16.8%	Recalibrate
Post	H6	extra_trees	54.4%	62.4%	53.6%	0.585	47.5%	Recalibrate
Post	H12	random_forest	58.3%	60.6%	56.7%	0.588	72.3%	Report
Validation	H3	random_forest	45.7%	40.9%	42.3%	0.643	25.7%	Recalibrate
Validation	H6	extra_trees	45.8%	46.1%	45.5%	0.632	13.2%	Recalibrate
Validation	H12	random_forest	42.2%	41.6%	40.8%	0.647	28.2%	Report

Validation accuracy is approximately 42%-46%, while post-disruption accuracy is approximately 54%-58%. The model carries useful directional information because its Brier score is lower than both baselines. However, a forecast can beat a baseline and still be poorly calibrated: predicted confidence does not consistently match observed frequency within every probability bin.

Classification behavior and overfitting diagnostic

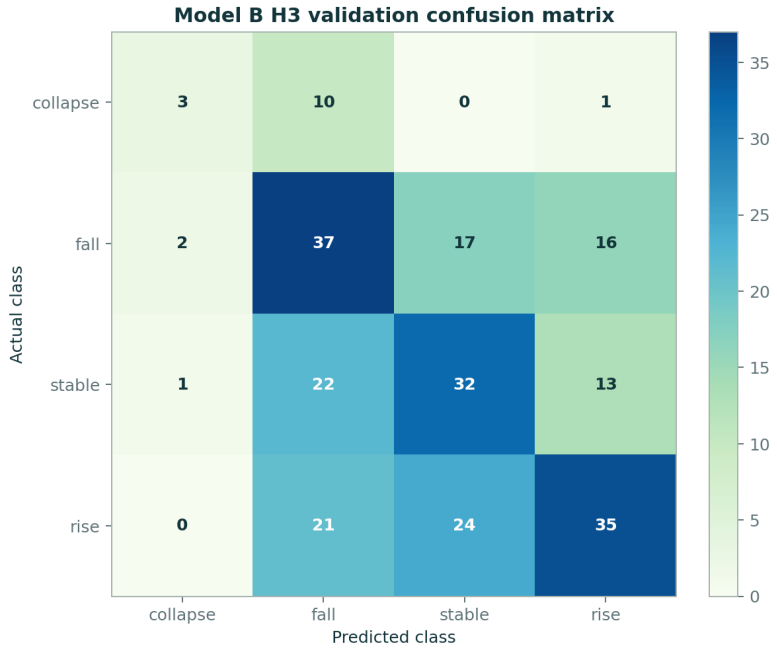


Figure 7. H3 blocked-validation confusion matrix. The model distinguishes common fall, stable and rise cases better than the relatively rare collapse class.

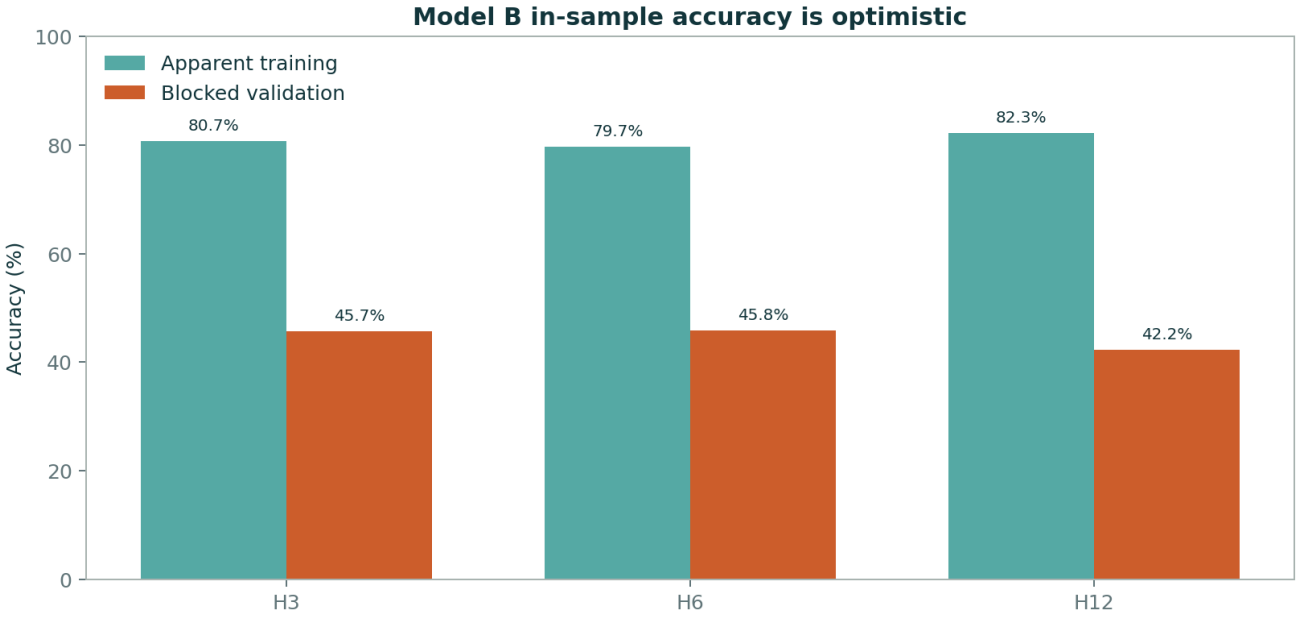


Figure 8. Apparent training accuracy is around 80%, materially above blocked-validation accuracy. This is an overfitting diagnostic; in-sample results do not determine demo readiness.

Disruption test, July 2021-June 2023: H3 accuracy 50.0%, Brier 0.603 and maximum gap 7.7%; H6 accuracy 47.2%, Brier 0.624 and maximum gap 8.3%. Both primary horizons pass. H12 has a 19.8% maximum gap and remains reporting-only.

Post-disruption monitoring: H3/H6/H12 accuracy improves to 56.1%/54.4%/58.3% and Brier scores remain below both baselines. Maximum gaps rise to 16.8%/47.5%/72.3%, so exact probabilities require recalibration and prospective monitoring.

8. Training, validation and test design

Both pipelines use blocked or rolling time splits. Random shuffling is not used.

Stage	Model A	Model B
Source history	Jan 2010-Jun 2026	Jan 2010-Jun 2026
Development pool	Origins through Jan 2022	Origins through Jun 2021
Blocked validation	Jan 2017-Jan 2022; 60 selected forecasts per HS/horizon	2018, 2019 and Jan 2020-Jun 2021 blocks
Disruption test	Feb 2022-Dec 2023; 23 origins per HS/horizon	Jul 2021-Jun 2023; 24 months x 6 suppliers
Post-disruption	Jan 2024-Dec 2025; 24 expected origins	Jul 2023-Jun 2026; realized labels shrink with horizon
Latest bundle origin	Jun 2026	Jun 2026
Training mode	Expanding/rolling, target availability enforced	Expanding/rolling, target availability enforced

For each simulated forecast origin, training includes only rows dated before the origin and only labels whose assumed publication date is already available. This blocks direct target leakage.

Model A uses separate pre-test residual histories to calibrate the selected model and the no-change baseline. This is important because sharing one calibration history would make MIS comparisons unfair.

Model B freezes the selected classifier family and temperature before the disruption test. Historical-frequency and last-observed-change baselines are scored on the same rows.

Evidence question	Model A	Model B
Does validation beat its baseline?	Yes: 6/6 configurations	Yes on Brier: 3/3 horizons
Are primary disruption gates met?	Yes: 4/4 H3/H6 combinations	Yes: H3 and H6
Is post-disruption behavior stable?	Mostly; 3/4 primary combinations pass	Brier remains useful; calibration fails
Can it be demonstrated?	Yes, with research-model label	Yes, as directional probabilities
Production approved?	No	No

9. Interpretation, limitations and next work

The models are useful evidence packages, but neither should be presented as production-approved.

Model A interpretation: the probabilistic intervals are the strongest result. Coverage is consistently high and the selected intervals are generally sharper than no-change. Median-price error is notably larger for HS 810890, so price-level forecasts for this code should be presented with the interval, not as a single precise number.

Model B interpretation: the model detects meaningful supplier-share direction better than simple probability baselines. The main weakness is calibration. A 60% predicted event should occur around 60% of the time; the post-disruption reliability bins do not consistently satisfy this.

Limitation	Effect	Required action
ERUC is not titanium-specific	Broad inflation may miss titanium market shocks	Seek a long titanium-specific benchmark when available
History starts in 2010	Model B class boundaries do not meet requested 2005 start	Keep deviation visible; extend only with authoritative data
Supplier quantities reconstructed	Model B inherits the aggregated shares in trade core	Rebuild from raw partner rows for production
Post-disruption data previously explored	Not a pristine final holdout	Collect a genuinely prospective holdout
Model B calibration drift	Exact probabilities may be over/under-confident	Recalibrate prospectively and monitor by horizon
H12 target incompleteness	Fewer independent outcomes and incomplete Model A post period	Treat H12 as reporting-only until more labels arrive
Model A source dependency missing	Selected V6 training script references an absent improved_v2 source file	Restore/vendor the dependency before CI or clean-room retraining

The saved Model A bundles, metrics, predictions and dashboards are present. However, the selected V6 training script imports a sibling improved_v2 Python source that is not currently inside the selected Model A folder tree. This does not invalidate the saved evidence, but it is a reproducibility and deployment-packaging blocker.

Recommended next sequence

- Restore the missing Model A inherited training source and run the package from a clean environment.
- Freeze the current artifacts and record checksums for data, model bundles and result files.
- Collect new prospective months without retuning; score them as a genuinely unseen holdout.
- Recalibrate Model B probabilities using only new validation history and recheck maximum calibration gap.
- Keep H12 reporting-only until the number of independent outcomes is sufficient.
- Define production monitoring thresholds for drift, missing sources, interval coverage and calibration.

10. Reproducibility and artifact map

The following files hold the training data, models and auditable predictions described in this report.

Artifact	Location within selected package
Model A training data	production_v6_titanium_core_corrected/data/...training.csv
Model A saved bundles	production_v6_titanium_core_corrected/models/*.joblib
Model A evaluation	production_v6_titanium_core_corrected/results/validation, stress_2022_2023, post_2024_2025
Model B training data	Training Titanium Model B/data/titanium_model_b_country_month_training_2010_2026.csv
Model B saved bundle	Training Titanium Model B/models/titanium_model_b_country_supply_shift_selected.joblib
Model B evaluation	Training Titanium Model B/results/*.csv and *.json
Combined evidence CSVs	Titanium Model A and B Performance Review/results
Combined dashboard	Titanium Model A and B Performance Review/visualizations/...dashboard.html

Metric interpretation rules

- Do not compare Model A MIS directly with Model B Brier score. They measure different tasks.
- Training diagnostics are secondary. Validation and time-ordered test results determine readiness.
- A pass means the implemented acceptance rule was met for that period. It does not mean the model is risk-free or production approved.
- A failed calibration gate does not mean classification has no value; it means the stated probability should not be treated as fully reliable.

Final assessment

Model A is the stronger demonstration package because its interval gates pass consistently across validation and the primary disruption period. Model B is a useful directional prototype that improves on simple probability baselines, but its validation and post-disruption probabilities need recalibration. The shared preprocessing is auditable and time-aware. Production approval remains pending prospective evidence, operational packaging and monitoring controls.